Lit roads, lighter crashes? Street lighting, motorcycles and crash severity in UK accident records

Group name	Team name
Dataset	UK road accidents, `accident_data.csv` (Kaggle upload of DfT STATS19 police records): kaggle.com/datasets/...
Code	github.com/.../cse558-midterm (Python scripts in `code/`, `README.md`, every reported number in `results/`)

#	Name	Roll no.	Institute email	Contribution
1	Name	Roll no.	email@iiitd.ac.in	Dataset, preparation (1–3)
2	Name	Roll no.	email@iiitd.ac.in	EDA, problem statements (4)
3	Name	Roll no.	email@iiitd.ac.in	Tests 1–2 (5)
4	Name	Roll no.	email@iiitd.ac.in	Tests 3–4, limitations, code (5, 6)

1. Dataset

Source: a Kaggle upload of the UK Department for Transport's STATS19 records of police-reported road accidents with at least one injury in Great Britain. One row is one accident. The accident IDs (e.g. `200701BS64157`) are STATS19 accident indices, with the year in the first four digits.

What is raw about it: the accident ID was turned into scientific notation (`2.01E+07`) for 239,884 rows, which creates 239,659 false duplicate IDs; for every readable 2007–2010 ID the date is exactly 12 years later than the ID year; six district names lost the letters "na" (`Stevege` for Stevenage); 19,739 cells are missing; and Fatal is only 1.3% of rows.

	Raw file	After encoding
Instances (rows)	670,679	670,660
Attributes (columns)	14	52
Instances × attributes	9,389,506	34,874,320

Size check: $670{,}660 \times 52 = 34{,}874{,}320 \geq 3{,}000{,}000$, so the requirement is met (the raw file alone gives $670{,}679 \times 14 = 9{,}389{,}506$). The 52 columns are 4 numeric, severity as an ordinal, the KSI flag, district frequency, and 45 one-hot columns: light 5, road surface 6, road type 6, urban/rural 3, weather 9, vehicle type 16.

Key features

Feature	Type	Real-world meaning
`Accident_Severity`	ordinal	worst injury: Slight, Serious or Fatal. Our response is **KSI** = Fatal or Serious (14.7% of crashes), the standard UK road-safety measure
`Light_Conditions`	nominal	daylight, or darkness with lights lit, unlit, none or unknown
`Urban_or_Rural_Area`	nominal	urban or rural location
`Road_Type`	nominal	single or dual carriageway, roundabout, one-way, slip road
`Road_Surface_Conditions`	nominal	dry, wet, frost/ice, snow, flood
`Vehicle_Type`	nominal	vehicle recorded for the crash, 15 types
`Number_of_Casualties`	discrete	people injured, 1 to 68
`Number_of_Vehicles`	discrete	vehicles involved, 1 to 32
`District Area`	nominal	local-authority district, 416 levels after cleaning

2. Data preparation

Issue found	What we did	Why, and not the alternative
19 exact duplicate rows	dropped	the same crash counted twice would add weight to it in every test
ID in scientific notation for 239,884 rows	kept the ID out of every key and every test	de-duplicating on the ID would have deleted 239,659 distinct crashes
dates 12 years ahead of the ID year (all 420,776 readable older IDs)	date not used, and dropped from the encoding	no test needs time; shifting back would require guessing for the IDs that are unreadable
6 district names missing "na" (3,438 rows)	merged back: `Stevege` → Stevenage, `Blaeu Gwent`, `North/South Larkshire`, `Clackmannshire`, `Crewe and ntwich`	Test 2 pairs crashes within districts; a split district halves its counts and adds a fake one
missing categories: weather 14,351, road type 4,580, surface 742, urban/rural 26, vehicle 7	recoded to an explicit `Unknown` level	filling with the mode would push 4,580 unknown road types into "single carriageway", which is PS1's filter; `Unknown` rows simply fall out of each filter
51 missing coordinates	kept as missing	coordinates are not used
casualty outliers: IQR fence is $Q_3 + 1.5\,\mathrm{IQR} = 1$	kept all values (max 68)	a fence at 1 would flag all 158,994 multi-casualty crashes; they are real (e.g. bus crashes). Test 4 is re-run without the 67 crashes above 10 casualties
vehicles above fence 3.5 (15,310)	kept	real pile-ups; PS2 fixes `Number_of_Vehicles` $= 2$ anyway
6 nominal columns	one-hot (45 columns); severity ordinal $0/1/2$ plus a KSI flag	categories have no order; KSI is the response
district, 416 levels	frequency-encoded	one-hot would add 415 sparse columns; the raw label is kept for Test 2

3. What changed after preparation

Measure	Before	After
Rows	670,679	670,660
Attributes	14	52 (encoded)
Missing cells	19,739	0 in the columns we use; 51 coordinates left, unused
Duplicate rows	19	0
Duplicate IDs	239,659	ID no longer used as a key
District levels	422	416 (6 typo levels merged)
Usable dates	0 (shifted)	dropped
KSI share	14.67%	14.67% (unchanged)

Turning severity into the binary KSI flag is what makes the proportion tests possible. Merging the district typos moved 3,438 crashes back to their real district, which changes the district-level pairs in Test 2. Only 19 rows were lost, so no group we test was thinned out by cleaning.

4. Basic questions and problem statement

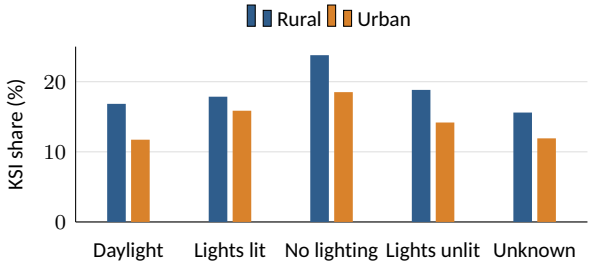


Figure 1: KSI share by light condition (all except Daylight are in darkness). Rural crashes on unlit roads are the most severe group: 23.8% KSI against 17.9% where lights are lit.

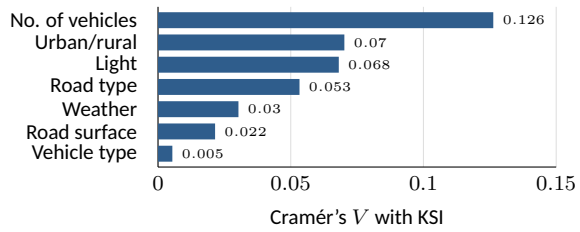


Figure 2: How strongly each column is associated with KSI. Vehicle type is the weakest by far ($V = 0.005$), 13 times weaker than light conditions. This warns us before PS2 is tested.

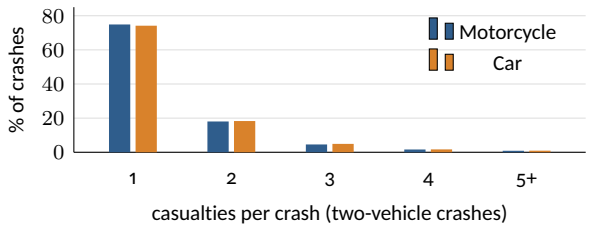


Figure 3: Casualties per crash are almost identical for motorcycles and cars: about 74% of crashes hurt exactly one person, and the counts are heavily right-skewed (skewness 7.0 over all crashes).

Basic questions

1. Which vehicle type is in the most crashes? Cars and taxis, in 77.4% of them.
2. Are rural crashes more severe than urban ones? Yes: 18.0% KSI against 12.8%.
3. Are crashes in the dark more severe than in daylight? Yes, in both areas (Figure 1).
4. Does bad weather make crashes worse? No: snowfall (9.7%) and icy roads (11.9%) have *lower* KSI shares than dry roads (15.1%), probably because people drive slower.

Our problem statements

PS1 (primary). Among rural single-carriageway crashes in darkness, is the KSI share higher where there is no street lighting than where lights are lit? Councils pay to light rural roads; if unlit crashes are no more severe, that spend is hard to justify on severity grounds.

PS2. In two-vehicle crashes, are motorcycle crashes more often KSI, while hurting *fewer* people per crash, than car crashes? If so, budgets split by casualty counts would understate motorcycle safety.

How they are novel: the basic questions compare one column against severity. PS1 compares two *darkness* groups on the same road type, so darkness itself is held fixed and only the lighting differs; it then checks the result within districts. PS2 asks two questions about the same groups, one on severity and one on casualty count, which could point in opposite directions. A plain "which vehicle has the most crashes" cannot show that.

5. Statistical inferences

Rows tested, PS1: 170,583 rural single-carriageway crashes, of which 28,201 were in darkness with no lighting (N) and 14,382 in darkness with lights lit (L). Excluded: "lights unlit" (608) and "lighting unknown" (1,683), whose lighting status is unclear, and daylight.

Rows tested, PS2: 397,950 two-vehicle crashes, of which 34,017 are motorcycles of any engine size (M) and 307,722 are cars or taxis (C). Other vehicles are excluded.

Test 1 (PS1): is the KSI share higher on unlit rural roads at night?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$p_N - p_L \leq 0$
$\boldsymbol{H_1}$	$p_N - p_L > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	two-proportion z-test (pooled SE for z, unpooled SE for the CI)
Smallest difference that matters	2 percentage points

p_N, p_L: KSI share of crashes on unlit and lit roads. One-sided, because the claim is that no lighting is worse. 2 pp means 20 more KSI crashes per 1,000 rural night crashes. That is roughly where the cost of a lighting scheme becomes worth discussing.

b) Other tests we considered

Two independent groups and a binary outcome, with expected cell counts in the thousands, call for a large-sample test on the difference of two proportions.

Also considered	Why we did not use it
Chi-square, 2×2	identical evidence ($\chi^2 = 65.32 = z^2$, $p = 6.4 \times 10^{-16}$ two-sided) but two-sided only, and it gives no CI for the difference
Fisher's exact test	needed only for small expected counts; our smallest is 3,348
CMH test by road surface	valid and stricter, since wet nights could confound. Run as a robustness check (below)
Logistic regression	adjusts for many confounders at once, but it is a model rather than one of our tests; left as an extension

c) Result

Estimate	$+3.50$ pp (24.46% unlit, 20.96% lit)
95% confidence interval	$+2.67$ to $+4.33$ pp
Test statistic (d.f.)	$z = 8.08$ (normal; χ^2 form has 1 d.f.)
p-value	3.2×10^{-16}
Decision	reject H_0

Rural night crashes on unlit roads are 3.5 pp more likely to be KSI, an odds ratio of 1.22. **It matters:** the whole CI lies above the 2 pp threshold. *Robustness:* stratifying by road surface (dry, wet, frost/ice, snow, flood), the CMH test gives $z = 9.61$, $p = 3.7 \times 10^{-22}$, with a Mantel–Haenszel risk difference of $+4.18$ pp, so road surface does not explain the gap. The gap is concentrated on dry (28.7% vs 23.6%) and wet (23.4% vs 19.3%) roads, and it disappears on frost or ice (14.4% vs 14.6%), where everyone already drives slowly.

Test 2 (PS1): does the gap hold within districts?

Test 1 treats every crash as independent, but crashes in one district share roads, policing and reporting habits. Here each district is its own control: for every district with at least 20 crashes in *both* groups we take $d_i = \text{KSI}_N - \text{KSI}_L$.

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_d \leq 0$
$\boldsymbol{H_1}$	$\mu_d > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	paired t-test on the district differences d_i
Smallest difference that matters	2 percentage points

b) Other tests we considered

Each district gives a pair of shares, so the test should use the within-district difference. The d_i look normal (skewness 0.10, Shapiro–Wilk $p = 0.72$), so a t-test on them is appropriate.

Also considered	Why we did not use it
Wilcoxon signed-rank	meant for skewed d_i or outliers, which we do not have. Run alongside: $W = 20{,}981$, $p = 2.3 \times 10^{-7}$
Sign test	valid, but uses only the sign of each d_i, so it has less power: 158 of 247 positive, $p = 6.7 \times 10^{-6}$
Unpaired Welch t-test	ignores the pairing and leaves district-to-district variation in the error: $t = 4.80$, $p = 1.1 \times 10^{-6}$, weaker than the paired $t = 5.57$ because the two shares are correlated across districts ($r = 0.26$)

c) Result

Estimate	mean $d = +3.08$ pp (median $+2.93$ pp; 247 of 396 districts, 37,189 crashes)
95% confidence interval	$+1.99$ to $+4.16$ pp
Test statistic (d.f.)	$t = 5.57$ (246)
p-value	3.3×10^{-8}
Decision	reject H_0

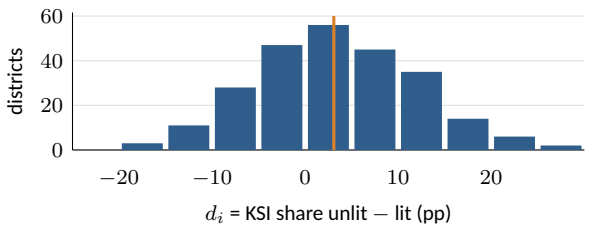


Figure 4: District differences d_i (247 districts). They are spread widely (SD 8.7 pp), but centred above zero; the orange line is the mean, $+3.1$ pp.

The within-district comparison gives the same answer as Test 1, so the result survives the clustering criticism. **It probably matters:** the estimate of 3.1 pp is above the threshold, but the lower end of the CI (1.99 pp) touches 2 pp, so a gap just below the threshold cannot be ruled out at district level.

Test 3 (PS2): are motorcycle crashes more often KSI?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$p_M - p_C \leq 0$
$\boldsymbol{H_1}$	$p_M - p_C > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	two-proportion z-test
Smallest difference that matters	5 percentage points

p_M, p_C: KSI share of two-vehicle crashes recorded with a motorcycle or a car. A gap below 5 pp would not justify a separate motorcycle-safety budget line.

b) Other tests we considered

Again two independent groups and a binary outcome with large counts.

Also considered	Why we did not use it
Chi-square, severity (3 levels) × group	uses all three levels, but the Fatal cells are thin and the result is harder to read. Supporting run: $\chi^2 = 1.65$ (2 d.f.), $p = 0.44$
Fisher's exact test	counts are far too large to need it (smallest expected count 301)
Relative risk with CI	useful alongside, but our threshold is an absolute difference: RR $= 0.986$ (0.956 to 1.017)

c) Result

Estimate	-0.16 pp (11.36% motorcycle, 11.52% car)
95% confidence interval	-0.52 to $+0.19$ pp
Test statistic (d.f.)	$z = -0.89$ (normal)
p-value	0.81
Decision	fail to reject H_0

There is no evidence that motorcycle crashes are more severe here. The CI is narrow and lies far below 5 pp, so this is a confident "no meaningful difference", not a lack of data. That contradicts national statistics, in which motorcyclists are heavily over-represented in KSI. Figure 2 explains why: in this file `Vehicle_Type` is almost unrelated to every other column (Cramér's $V \leq 0.012$ with severity, area, light, road type and number of vehicles), and the vehicle mix in single-vehicle crashes matches that in multi-vehicle crashes to within 0.1 pp (e.g. buses 3.93% vs 3.92%). The column seems to record one vehicle per accident, not the vehicle of the person who was hurt.

Test 4 (PS2): do motorcycle crashes hurt fewer people?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_M - \mu_C \geq 0$
$\boldsymbol{H_1}$	$\mu_M - \mu_C < 0$
Significance level	$\alpha = 0.05$
Side	one-sided (left)
Test	Welch t-test (unequal variances)
Smallest difference that matters	0.10 casualties per crash

μ_M, μ_C: mean casualties per two-vehicle crash. 0.10 means 10 fewer injured people per 100 crashes, enough to change ambulance and hospital load.

b) Other tests we considered

Two independent groups and a mean of counts. With n in the tens of thousands, the CLT makes the sample means normal despite the skew (Figure 3), and σ is unknown, so we use a two-sample t-test. We chose Welch's version because the spreads differ.

Also considered	Why we did not use it
Pooled t-test	assumes equal variances, rejected by Levene ($W = 11.0$, $p = 0.0009$) and by the F-test ($F = 0.93$, $p = 10^{-17}$). Run anyway: $t = -3.32$, $p = 0.0005$, same decision
Two-sample z-test	needs the population σ; at this n it gives the same $z = -3.41$, $p = 0.0003$
Mann–Whitney U	74% of values are tied at 1, and it tests a shift in distribution, not the mean. Robustness run: $p = 0.001$, $P(M > C) + \frac{1}{2}P(\text{tie}) = 0.496$, practically a coin flip
Poisson rate test	casualties start at 1, and their variance is *below* the mean (var/mean $= 0.44$ for M, 0.47 for C); shifted to start at 0 they are overdispersed (1.65, 1.70). Either way the Poisson variance is wrong; it gives $p = 0.012$

c) Result

Estimate	-0.015 casualties per crash (1.362 motorcycle, 1.377 car)
95% confidence interval	-0.024 to -0.006
Test statistic (d.f.)	$t = -3.41$ (42,493)
p-value	3.2×10^{-4}
Decision	reject H_0

Motorcycle crashes do hurt fewer people on average, but **it does not matter**: the whole CI lies inside ± 0.10, and the gap is 1.5 people per 100 crashes. With $n = 341{,}739$, a negligible gap is still significant. Dropping the 67 crashes with more than 10 casualties changes nothing (-0.015, $p = 3.3 \times 10^{-4}$).

PS2 answer, and the multiple-testing check

The expected tension between Tests 3 and 4 does not appear: motorcycle crashes are neither more severe nor meaningfully smaller. Given Figure 2, we read this as a property of how `Vehicle_Type` was recorded, not as evidence that motorcycles are safe. Across the four tests a Holm correction (adjusted p: 1.3×10^{-15}, 9.8×10^{-8}, 0.81, 6.5×10^{-4} for Tests 1–4) **changes no decision**.

6. Limitations

- **No speed-limit column (the main confounder for both statements).** In Great Britain a road with street lighting defaults to 30 mph, while an unlit rural single carriageway defaults to 60 mph. Part of the 3.5 pp gap in PS1 may be speed, not light, and PS2 cannot be adjusted for speed either.
- **Association, not cause.** Lights are not placed at random: they go on junctions and built-up stretches. Only a before-and-after study of newly lit roads could show cause.
- **`Vehicle_Type` looks detached from the crash.** Its near-zero association with every other column means PS2's conclusions describe this file, not motorcyclists in Great Britain.
- **Injury crashes only.** STATS19 records crashes with at least one injury that were reported to police, so the KSI share is a share of injury crashes, and slight injuries are under-reported.
- **Districts kept.** Test 2 uses 247 of 396 districts; the 149 with fewer than 20 crashes in a group are left out, and the result may not apply to them.
- **Shifted dates.** Without usable dates we cannot check seasonal confounding (dark evenings in winter coincide with wet roads); the CMH check by road surface covers only part of this.